

TriZOD — Project Update

LACS Integration, Duplicate Entries & BindBox Comparison

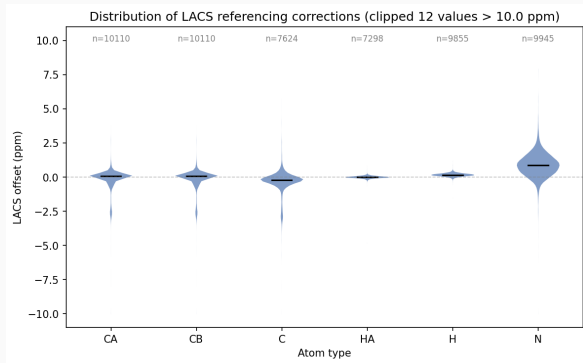
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TriZOD Project Meeting

LACS Integration Results

LACS Offset Distribution Across the BMRB



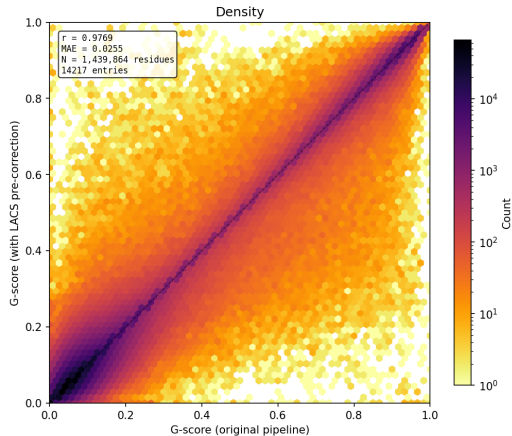
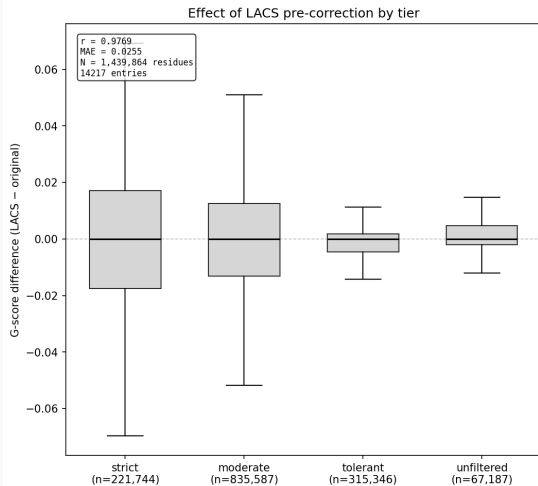
LACS offsets computed for **~10,000 BMRB entries** per atom type.

Key observations:

- CA/CB: tightly centered, ~20% show >0.5 ppm offsets
- C': widest ^{13}C spread (up to ± 5 ppm)
- HA/H: very narrow (^1H scale)
- **N: largest offsets**, broad distribution, systematic positive bias (~35% mis-referenced)

12 entries with offsets >10 ppm clipped.

Effect of LACS Pre-Correction on G-Scores



Duplicate Entry Analysis

Same Protein, Multiple Experiments

5,393 entries across **1,939 unique sequences** with ≥ 2 BMRB entries.

Matching criteria: temperature within 10 K, pH within 1.0.

Protein	Len	Entries	Bound
α -synuclein	140	62	1
human TSG-6	98	33	18
Protein-export protein SecB	155	28	0
AbpSH3	59	25	0

Two questions:

1. How reproducible are independent measurements of the *same* state?
2. Can we detect **binding-induced** shift changes above the noise floor?

Bound vs. Unbound Shift Differences

581 bound/unbound pairs compared (mean absolute shift difference):

Atom	Pairs	Mean abs	Median abs	>0.5 ppm	>1.0 ppm
CA	398	0.481	0.330	22.5%	12.3%
CB	384	0.532	0.349	24.5%	12.2%
C	246	0.399	0.231	19.1%	10.2%
HA	299	0.096	0.057	3.6%	0.8%
H	542	0.139	0.079	5.9%	1.7%
N	492	0.763	0.453	32.8%	19.0%
HB	295	0.088	0.048	2.7%	0.6%

N shows largest binding effects (median 0.45 ppm), consistent with backbone sensitivity to ligand binding. ^1H atoms show much smaller perturbations.

Measurement Reproducibility & Binding Signal

Independent measurements (1,301 pairs):

Atom	Mean abs	Median abs
CA	0.298	0.206
CB	0.305	0.205
C	0.547	0.472
HA	0.045	0.030
H	0.069	0.049
N	0.356	0.229
HB	0.101	0.056

This is the noise floor — variation between independent measurements of the same protein in the same state.

Binding signal / noise ratio:

Atom	Binding	Noise	Ratio
CA	0.481	0.298	1.6×
CB	0.532	0.305	1.7×
C	0.399	0.547	0.7×
HA	0.096	0.045	2.2×
H	0.139	0.069	2.0×
N	0.763	0.356	2.1×
HB	0.088	0.101	0.9×

Ratio > 1: binding effect exceeds noise.

C and HB are below noise — not reliable binding indicators at this scale.

BindBox Comparison

BindBox: Interactive NMR Shift Analysis

BindBox (Bind Research, UK) — Streamlit web app for IDP chemical shift analysis.

Main tools:

1. **BMRB Dashboard** — 1D/2D/3D shift distributions, Gaussian fitting, POTENCI overlay
2. **SpinForecast** — Bayesian peak assignment without sequential connectivities
3. **POTENCI** — same algorithm as TriZOD (Nielsen & Mulder 2018)
4. **pH/T Adjustment** — shift prediction under changed conditions

Their dataset:

- 3,320 BMRB entries (disordered regions, AlphaFold2 pLDDT < 70)
- 480,599 shift rows, pre-computed Parquet files
- BMRB September 2025 release

Preprint in preparation (~May 2026): probabilistic IDP peak assignment.

BindBox vs. TriZOD

Aspect	BindBox	TriZOD
Interface	Web GUI (Streamlit)	CLI + Python library
Mode	Interactive, single-protein	Batch, database-scale
Scale	3,320 entries	17,388 entries
Primary output	Visualizations, probabilities	Per-residue Z/G-scores
Data storage	Pre-aggregated Parquet	Raw NMR-STAR + caches
Filtering	User-driven UI controls	Systematic 4-tier (16 criteria)
Referencing	User-supplied offsets	LACS automatic detection
Peak assignment	SpinForecast (Bayesian)	Not in scope
Disorder scoring	Not in scope	Core purpose
Disorder def.	AlphaFold2 pLDDT	CheZOD Z/G-score

Complementary, not competitive: BindBox provides interactive single-protein exploration; TriZOD provides systematic database-scale scoring.